Aniket Pal Sec A Roll 1 BTech(CSE - Networks)

Data Sheet

EXPERIMENT NO. 07

Title: - Simulation of Differentiator and Integrator Circuit using Op-Amp (using Tinkercad/ Proteus software circuit).

Instrument & Component Required for Differentiator: -

Sl. No.	Apparatus / Component	Specification	Quantity
1.	OP - AMP		1
2.	Resistance		
3.	Capacitor		
4.	Connecting Wires		
5.	Ground		
6.	Function Generator & Oscilloscope		

Observation Table for Differentiator: -

I/P Waveform	P-P Amplitude of I/P (V)	I/P Time Period (ms)	P-P Amplitude of O/P (V)	O/P Time Period (ms)
Sine	1.0	10	0628	10
Sine	1.5	10	0.942	10

Instrument & Component Required for Integrator: -

Sl. No.	Apparatus / Component	Specification	Quantity
1.	OP-Amp		
2.	Resistance & Capacitor		
3.	function Generator		
4.	Oscilloscope		
5.	Connecting Wires		
6.	Ground		

Observation Table for Integrator: -

I/P Waveform	P-P Amplitude of I/P (V)		P-P Amplitude of O/P

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Title :- Simulation of Differentiator and Integrator Circuit Using OP-Amp (Using Tinkercad / Proteus Circuit).

Theory :- By replacing Standard feedback resistors With Capacitors, operational amplifiers With Capacitors, operational amplifiers Can perform real-time calculus operations on time-varying input Signals. An integrator Circuit accumulates the input Signal over time, outputting a voltage proportional to the mathematical integral of the input wave: $V_o = -(1/R_1C_f) \times \int V_i dt$.

Conversely, a differentiator Circuit produces an output voltage that is directly proportional to the rate of change of the input signal:

$$V_o = -R_f \times C_1 \times (dV_i/dt).$$

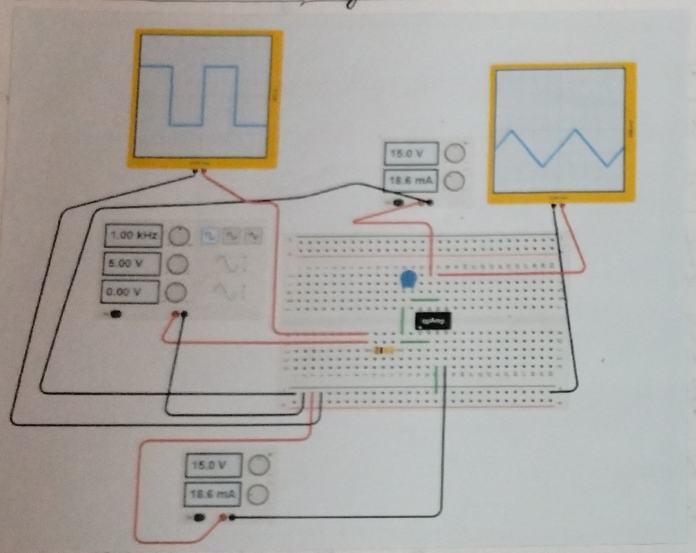
For Differentiator:-

Component Required:-

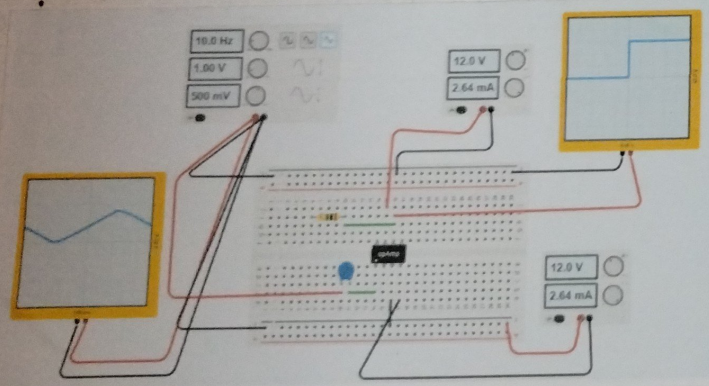
Sl.No	Components	Specification	Quantity
1.	OP-AMP	?	?
2.	Resistance		
3.	Capacitor		
4.	Connecting Wires		1
5.	Ground		
6.	Function Generator & Oscilloscope		

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Differentiator Circuit Using OP-Amp:-



Integrator Circuit Using OP-Amp:-



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Observation Table:-

I/p Waveform	(P-P) Amplitude of I/P (V)	I/P Time Period (ms)	(P-P) Amplitude of O/P (V)	O/P Time Period (ms)
Sine	1.0	10	0.628	10
Sine	1.5	10	0.942	10

For Integrator:-

Sl.No	Components	Specification	Quantity
1.	OP-Amp		
2.	Resistance & Capacitor	?	?
3.	Function Generator		
4.	Oscilloscope		
5.	Connecting Wires		
6.	Ground		

Observation Table:-

I/P Waveform	(P-P) Amplitude of I/P (V)	I/P Time Period (ms)	P-P Amplitude of O/P (V_o)	O/P Time Period (ms)
Square	1.00	10	0.50	10
Square	1.50	10	0.75	10

Calculation:-

Differentiator Output, $V_o = -R_f C_1 (dV_i/dt) = 0.50$ V (P-P)

Integrator Output, $V_o = -(1/R_1 C_f) \int V_i dt = 0.75$ V (P-P)

Conclusion:-

The handwritten data sheet left the conclusion blank. The recorded voltage amplitudes demonstrate that the circuits successfully functioned as active filters, performing continuous differentiation and integration on the supplied waveforms.

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